West Visayas State University
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
Luna St., La Paz, Iloilo City 5000
Iloilo, Philippines
* Trunkline: (063) (033) 320-0870 loc 1403 * Telefax No.: (033) 320-0879
* Website: www.wvsu.edu.ph * Email Address: cict@wvsu.edu.ph

Unit 8 Assignment

Name:Luisa Rose D. Brillantes Date: 11/12/26

Year and Section: BSIT 3-A

Note: You don't need to upload your Flutter project on this assignment.

I. CICT Org Fair App Development part 4. The Org Fair app works on your computer. On fair day it has to work on the phones of 300 students who never saw your code. In this assignment you **release** it: a live website anyone can open, an Android app testers can install, and a pipeline that tests and deploys every change for you. You will also see the difference between changing **content** (the CMS: no release needed) and changing **code** (a new release, and maybe a rollback).

Please do this preliminary steps before working on your

1. Integrate your existing CMS, used in your previous assignment. If you have used a local Strapi server, you need to move it to a host (Strapi Cloud).
2. Use the Flutter project from your Unit 7 assignment. Point it at your CMS with a cms.json file (copy cms.example.json). Please see this link as reference on setting up.
3. Make your own public GitHub repository, for example org-fair (separate from the class submission repository: GitHub Pages and Actions need a repository you own). Commit everything, including web/ and android/. The given .gitignore keeps cms.json, key.properties and keystores out. Run git status and check that cms.json is not listed.

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Refer to the table below on the files you need to add in your current Flutter Project, use this GitHub Repo as a reference:

New in the starter	What it is	What you do
`lib/screens/about_screen.dart`	VersionTile reads the version with package_info_plus	TODO A2
`.github/workflows/ci.yml`	CI: analyze, test and build on every pull request and push	Read
`.github/workflows/deploy.yml`	CD: test, build and publish to GitHub Pages on every push to main	TODO C1, C2
`cms.example.json`	The --dart-define values; copy it to cms.json	Copy and fill in
`.gitignore`	Keeps secrets and build output out of Git	Read

Do the following steps to accomplish this assignment:

A. Prepare the release (12 points)

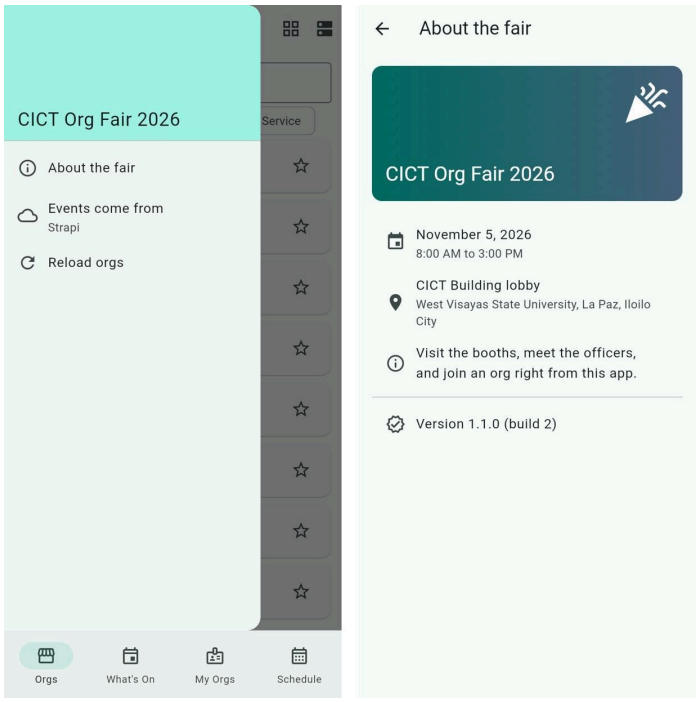
a. **(2 points)** In pubspec.yaml, change version: 1.0.0+1 to 1.1.0+2.

b. **(3 points)** Finish the TODO in VersionTile so the About screen shows "Version 1.1.0 (build 2)"

c. **(3 points)** Give the app its real name: <title> and apple-mobile-web-app-title in web/index.html, name, short_name and description in web/manifest.json, and android:label="CICT Org Fair" in android/app/src/main/AndroidManifest.xml.

d. **(2 points)** Add the internet permission to the main Android manifest, inside <manifest> and above <application>. Debug builds have it already; release builds do not, and the app would show "No connection".

<uses-permission android:name="android.permission.INTERNET"/>



The deployed build: the drawer names the CMS, and About shows the version from pubspec.yaml (read from version.json on the web).

e. **(2 points)** Screenshot git status (or your repository on GitHub) showing that cms.json is not committed.

Screenshot:

B. Deploy to the web **(20 points)**

a. **(3 points)** Build: flutter build web --release --dart-define-from-file=cms.json. Open build/web/version.json. What does it contain, and which screen of your app reads it?

Answer:

b. **(5 points)** Deploy with **Firebase Hosting** or with **GitHub Pages** through Part C. Put the live URL at the top of your README.

c. **(4 points)** Open the live site on your phone and take a screenshot of What's On. If the events do not load but your **CMS** URL works in a browser tab, open DevTools > Console: a CORS error means the CMS does not trust your site's address yet (Sanity: add it under API > CORS origins). An https site also cannot call an http CMS. Write down any problem you hit and how you fixed it.

Answer (if any):

d. **(4 points)** Content without a release. In the CMS, cancel one event and change another one's venue. Reload the live site. Screenshot before and after. You did not rebuild or redeploy anything: explain why it still changed.

e. **(4 points)** Search build/web/main.dart.js for your CMS address and token (Ctrl+F, or grep -c "your-value" build/web/main.dart.js). Write what you found, and why that is acceptable only for a read-only token.

Answer:

C. CI/CD with GitHub Actions **(25 points)**

a. **(3 points)** In deploy.yml, TODO C1: run the workflow on every push to main, and by hand (workflow_dispatch).

b. **(5 points)**TODO C2: build with --base-href "/${{ github.event.repository.name }}/" (GitHub Pages serves your site from /<repository>/) and pass each CMS value with --dart-define, reading them from repository variables and the token from a secret.

```
run: >-
  flutter build web --release
  --base-href "/${{ github.event.repository.name }}/"
  --dart-define=CMS=${{ vars.CMS }}
  --dart-define=CMS_URL=${{ vars.CMS_URL }}
  ...the other values the same way...
  --dart-define=CMS_TOKEN=${{ secrets.CMS_TOKEN }}
```

c. **(5 points)** On GitHub: Settings > Pages > Source: GitHub Actions. Settings > Secrets and variables > Actions: add the variables (CMS, CMS_URL, ... whatever your CMS needs) and, only if needed, the secret CMS_TOKEN. Push. In the Actions tab, open the run and click the deployment link https://<user>.github.io/<repository>/.

d. **(8 points)** CI catches a bug. Make a branch, change the chip text "Cancelled" to "Canceled" in event_tile.dart, push and open a pull request. Screenshot the red X and the failing test. Fix it on the same branch, screenshot the green check, merge, and watch the deploy run start by itself.

e. **(4 points)** Is your deploy.yml continuous delivery or continuous deployment? What would you change to make it the other one?

D. Android release **(15 points)**

a. **(3 points)** Create an upload keystore outside your project folder. Remember the passwords. For educational purposes only, take a screenshot of your passwords and put it here.

Command:

```
keytool -genkey -v -keystore ~/upload-keystore.jks -keyalg RSA -keysize 2048 -validity 10000
-alias upload
# Windows (PowerShell):
keytool -genkey -v -keystore $env:USERPROFILE\upload-keystore.jks -storetype JKS -keyalg RSA
-keysize 2048 -validity 10000 -alias upload
```

Screenshot:

b. **(5 points)** Create android/key.properties (it is in .gitignore) and add the signing config to android/app/build.gradle.kts.

Command:

```
storePassword=your-store-password
keyPassword=your-key-password
keyAlias=upload
storeFile=/home/you/upload-keystore.jks     (Windows: C:\\Users\\you\\upload-keystore.jks)
```

build.gradle reference:

```
// android/app/build.gradle.kts (from docs.flutter.dev/deployment/android)
import java.io.FileInputStream
import java.util.Properties
// ^ the imports go at the very top of the file

val keystoreProperties = Properties()
val keystorePropertiesFile = rootProject.file("key.properties")
if (keystorePropertiesFile.exists()) {
    keystoreProperties.load(FileInputStream(keystorePropertiesFile))
}

android {
    // ... keep what flutter create wrote ...
    signingConfigs {
        create("release") {
            keyAlias = keystoreProperties["keyAlias"] as String
            keyPassword = keystoreProperties["keyPassword"] as String
            storeFile = keystoreProperties["storeFile"]?.let { file(it) }
            storePassword = keystoreProperties["storePassword"] as String
        }
    }
    buildTypes {
        release {
            signingConfig = signingConfigs.getByName("release")
        }
    }
}
```

c. **(4 points)** Build: flutter build apk --release --dart-define-from-file=cms.json. The file is build/app/outputs/flutter-apk/app-release.apk. Install it on a phone (flutter install, or copy the file) and check that “What's On” loads and the About screen shows 1.1.0 (build 2).

   Screenshot:

d. **(3 points)** Why must every update be signed with the same key? What happens if you lose upload-keystore.jks with Play App Signing, and without it?

   Answer:

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Finally, provide the following URLs for evidence:

1) GitHub repository:

2) Deployed Web App URL: